PILOT PARTNERSHIP · STUDY CLUB

Study Club
AI Engineer

VETERANTECH × NETRA RUNTIME PILOT

An eight-week online class in which 8 teams of selected mentees build and deploy AI apps on Netra inference. Veterantech runs the class, the mentors and the gateway that every request passes through.

8 WEEKLY SESSIONS · ONLINE

24–30 MENTEES · 8 TEAMS

8 LIVE APPS ON NETRA

START DATE TO BE AGREED

PREPARED BY

KSM Veterantech, Faculty of Computer Science, UPN "Veteran" Jakarta

PREPARED FOR

Netra Runtime, after the partnership call on 31 October 2026

01 · OVERVIEW

Eight weeks. Eight live AI apps on Netra.

Veterantech proposes a pilot AI Engineer class inside its Study Club program. Netra provides the inference. Veterantech runs the class, the mentors and the gateway. This document sets out what each side does, what gets delivered, and what still needs an answer before week 1.

THE CLASS — PROPOSED

WHAT	**An 8-week AI Engineer track.** Mentees build production-ready AI apps. It is not an intro to programming.
WHO	**24–30 selected mentees** in 8 teams of 3–4. Backend and Data Science alumni first, then open screening.
FORMAT	One 120-minute online session a week with 1–2 mentors.
OUTPUT	**8 apps live on Netra**, with 6 as the minimum.

THE PARTNERSHIP — FOR AGREEMENT

NETRA	**Credits and models**, starting with a pool sized after week 2, **its name on the cohort**, and Netra on stage.
VETERANTECH	The class and mentors, the gateway, **two usage and feedback reports**, public apps and a showcase that tags Netra.
CONTROL	Mentees never hold a Netra key. Every call passes Veterantech's gateway.
WHEN	Two months. The start date follows confirmation of credits and models.

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02 · WHO DELIVERS IT

Veterantech has shipped 15+ products since 2014.

Veterantech is a student-led technology organization under the Faculty of Computer Science at UPN "Veteran" Jakarta. Members learn in weekly study clubs, enter competitions, and build products for real users and clients.

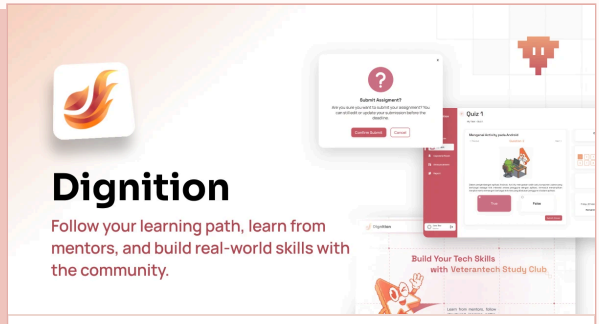
15+	Rp300M+	2014	10+
PRODUCTS DEVELOPED	TOTAL PROJECT VALUATION	OPERATING SINCE	COMPETITION WINS



siera.veterantech.id

SIERA

Built in-house for the university. SIERA moved UPNVJ's 2026 new-student orientation online for **4,900+ users**, with automated attendance validation, task management and live monitoring.



dignition.veterantech.id

Dignition

Our own learning platform already runs the Study Club: **740+ registered learners, 22 mentors, 19 study clubs** and 217 learning resources. Its MCP connector works from Claude, ChatGPT and Cursor.

The Study Club runs six tracks today: Frontend, Backend, UI/UX, Data Science, Android Dev and Product Management. AI Engineer becomes the next track and takes its fast-track mentees from Backend and Data Science. Vexpro, our client-project program, is where "Netra first" applies (section 09).

Figures: Kabinet Genesis profile deck; SIERA totals from Veterantech (31 Oct 2026); Dignition public stats endpoint (30 Oct 2026), counting registered accounts, not active users.

03 · THE COHORT

Selected builders, not a beginner class.

The class takes mentees who already build backends or work in Python. Selection keeps Netra's credits with people who will spend them on real builds.

24–30	8	8 × 120	1–2
SELECTED MENTEES	TEAMS OF 3–4	WEEKLY SESSIONS, MINUTES	MENTORS PER SESSION

SELECTION PATHS

PATH	WHO	WHAT WE VERIFY	PRIORITY
Fast-track Backend	Alumni of Study Club Backend Basic	HTTP and REST APIs, TypeScript or JavaScript, server setup, `.env`, JSON and async code	First
Fast-track Data	Alumni of Study Club Data Science Basic	Python and ML reasoning, plus a passed test of calling a REST API outside Jupyter	Second
Open screening	Non-alumni with a portfolio	Python, ML reasoning and API connections, shown through past work	Selective

FORMAT

- One live session a week on Zoom, 120 minutes (90 at minimum)
- Each session pairs a lesson with a hands-on lab
- Teams form at the end of week 1; each team builds one AI app
- Veterantech provides starter kits so labs start from working code

DROP-OUT RULE

- More than 3 unexcused absences, or missing assigned work, ends a mentee's place in the class
- Their gateway key is revoked the same day
- The credit they would have used stays with the class

What each mentee gets. A personal gateway key, starter kits for every lab, weekly mentor feedback, a public app with their team, and a Certificate of Competence on passing (section 05).

Cohort, selection and format: Veterantech academic team, Study Club AI Engineer plan (Kabinet Genesis).

04 · CURRICULUM

Eight sessions, each ending in something that runs.

Weeks 1–4 are individual exploration. Weeks 5–8 are team projects that end on a public server. Every lab calls Netra through the gateway.

WK	MODULE	WHAT MENTEES BUILD	MILESTONE
P1	Setup and persona	Netra calls through the gateway with streaming; handling 401, 429 and 5xx with retry and timeout; a system prompt and persona; basic hardening (`.env`, input validation, `max_tokens`, CORS)	Gateway connected; teams formed
P2	Cost and routing	Cost per request from logs; structured JSON with schema validation; a semantic cache; a model router with fallback; golden set v1 of 15 cases	Assignment 1 released (solo)
P3	Tools and MCP	Function calling; an agent loop with an iteration limit; human confirmation for a risky tool; a local MCP server with 2–3 logged tools	Tools running; team proposal
P4	Vectors and RBAC	Embeddings in a managed vector DB; chunking and metadata; role filters set by the server, never by the user or the model; leak tests	Role-based search; Assignment 1 due
P5	Secure RAG	Ingest, retrieve and generate with citations; hybrid search and reranking; RAG evaluation with recall@k and faithfulness	Assignment 2 released (team)
P6	Guardrails	OWASP Top 10 for LLM apps; input and output guards; indirect prompt injection; PII redaction; cross-team red-teaming with 15–20 attacks	Attack success rate measured
P7	Observability and deployment	Request tracing and tool-call audit; p95 latency, error and cost metrics; Docker, HTTPS, health checks and rollback on a team server	App live on a server; Assignment 2 due
P8	Demo day	8–10 minute team demos with metrics: golden-set pass rate, RAG faithfulness, attack success rate, cost per query, p95 latency	Final submission; Netra judging

Depends on Netra. Weeks 3–5 need tool calling and an embedding model. Both are part of ask 01 in section 08.

Netra on stage. We propose the guest session for week 7, where observability and deployment meet the provider side.

Curriculum owner: Veterantech academic team. Module content may be refined before week 1 without changing the milestones.

05 · DELIVERABLES AND GRADING

What mentees ship, and how mentors grade it.

Three graded hand-ins lead to one public app per team. The final project carries half the grade, and its rubric rewards security and evaluation as much as features.

ASSIGNMENT 1 · SOLO — WEEKS 2 → 4

A smart gateway client

Structured output, semantic cache, model router and app-level rate limit, run against golden set v1, with a short report.

ASSIGNMENT 2 · TEAM — WEEKS 5 → 7

A secure RAG app

RBAC, citations and an evaluation report of at least 20 questions. It becomes the base of the final project.

FINAL PROJECT · TEAM — WEEK 8

The submission package

Code, README, metrics report, a link to the live app and a short demo video.

GALLERY · PUBLIC — WEEK 8

8 apps live on Netra

Each team's app runs on a public server with an open-source repo that credits Netra. The minimum is 6.

FINAL GRADE

COMPONENT	WEIGHT
Assignment 1 (solo)	15%
Assignment 2 (team)	25%
Final project (team)	50%
Attendance and participation	10%

FINAL PROJECT RUBRIC

CRITERION	WEIGHT
Functionality and product value	25%
Security, RBAC and guardrails	25%
Evaluation, metrics and observability	20%
Architecture and code quality	15%
Presentation and live demo	15%

A Certificate of Competence goes only to mentees who meet attendance, hand in every assignment and score above 80.0. With Netra's approval, the certificate reads "Study Club AI Engineer, powered by Netra". Mentors adjust individual scores within a team using git commit history and peer review.

06 · GATEWAY AND ACCESS CONTROL

Mentees never hold a Netra key.

Every request goes through Veterantech's LLM gateway, built on LiteLLM and Postgres. It checks the mentee's key, limits and budget, forwards the call to Netra with the one real key, and logs usage per request.

01 · MENTEE

Mentee's code

Standard OpenAI SDK, our base URL and a personal key.

02 · VETERANTECH

LLM gateway

Checks key, limits and budget. Logs tokens, cost and timing. Holds the Netra key in its server environment.

03 · NETRA

Netra API

One class key, held by Veterantech only.

RUNNING TODAY — BUILT AND TESTED

- A personal virtual key per mentee, expiring after 90 days
- Per-key US$ budgets (daily and monthly) and rate limits (requests and tokens per minute)
  Cap levels for the class will be agreed with Netra.
- Only the models we allow for each key
- Tokens, cost, start, first-token and end time logged per request
- One call blocks one key; nobody else is affected

Tested: an empty budget returns HTTP 422, a blocked key 401, a model outside the key's list 403.

PLANNED BEFORE WEEK 1 — NOT BUILT YET

- Input caps of 4,000 tokens in weeks 1–4 and 8,000 in weeks 5–8, for RAG context
- A circuit breaker that blocks a key after more than 20 error responses in one minute
- A `VT-AI-` prefix on mentee keys
- Class rate limits per mentee, raised for live sessions, at levels agreed with Netra

Waiting on Netra. The gateway answers with a mock model today. It switches to Netra once we have a key, base URL, model IDs and per-token prices. Without prices, every call costs $0 and budgets never block.

Student data. Gateway records carry an opaque mentee ID, never a name or student number. The request log stores usage, not prompts.

Known limit: spend reaches the database in batches about 15–20 seconds after a request, so a budget is a close cap rather than a real-time wall. Parallel calls cannot overrun it, because the gateway reserves each request's worst-case cost on arrival.

07 · CREDITS AND USAGE

A starting pool, sized together after week 2.

We do not fix a token number up front. Netra opens a starting pool for the first weeks, and after week 2 we size the pool for the rest of the pilot together, from real gateway usage. We put no money value on it until Netra confirms the models and their prices.

01 · START — WEEK 0

Netra opens a starting pool

Enough for setup, the first live labs and Assignment 1, on the models we agree.

02 · MEASURE — WEEKS 1–2

The gateway logs real use

Tokens per mentee, team, session and model, peak requests per minute in live labs, and cost per request once prices are set.

03 · SIZE — AFTER WEEK 2

We size the pool together

Netra and Veterantech set the pool for weeks 3–8 from those numbers. The week-4 review can adjust it again.

TRAFFIC PATTERN

- **Peak.** Live hands-on labs: about 30 mentees for 45 minutes. This is when the class needs bursts up to 20 RPM.
  Burst capacity to be confirmed with Netra's technical team.
- **Regular.** Independent work and team projects, spread outside class hours at a low, even load.

WHAT CHANGES USE OVER THE PILOT

- **Weeks 1–4.** Individual labs and Assignment 1, mostly short calls.
- **Weeks 5–6.** RAG with larger contexts and red-teaming run the highest.
- **Weeks 7–8.** Eight team apps live on public servers, then demo day.

Keeping to the pool. Once the pool is agreed, we track projected use against it each week. If projected use passes 80% of the pool, we lower daily limits or move labs to a smaller model. The week-4 review with Netra can also adjust models or scope.

Indicative floor, an early assumption from the academic team's plan that will be revised with real usage: 30 mentees × 1.5M tokens (weeks 1–4) + 8 teams × 5M (weeks 5–8) + 15M buffer. Agent loops and 8,000-token RAG contexts will likely push real use above it. Money value: set from Netra's prices for the agreed models.

08 · WHAT NETRA PROVIDES

Three things the class needs from Netra.

Three asks, each tied to the weeks that depend on it. The operating details below let the gateway switch from the mock model before week 1.

01 **Credits and the models the class needs** — FROM WEEK 1

A starting pool sized together after week 2, on models with tool calling and embeddings. Education credits work too. Weeks 3–5 depend on tool calling and embeddings.

02 **Your name on the cohort** — WEEKS 1–8

We run it as "Study Club AI Engineer, powered by Netra", with Netra on every graduate's certificate. This needs Netra's permission to use its name and logo for the cohort.

03 **Netra on stage** — WEEKS 7 AND 8

A guest session and a judge's seat at demo day, with the teams whose apps run on Netra. Demo-day format, online or on site, is still open.

SETUP AND OPERATING REQUIREMENTS BEFORE WEEK 1

ITEM	WHY THE CLASS NEEDS IT
API key for the class	Held only in the gateway's server environment; mentees never see it
Base URL	The gateway forwards every mentee call to it
Model IDs, chat and embedding	Each mentee key allows only the agreed models
Per-token prices per model	Budgets and cost logs depend on them; without prices nothing blocks
Maximum output tokens per model	The gateway reserves each request's worst case against the budget
Rate limits on the class key	Live labs put about 30 mentees on it at once, so the class needs bursts up to 20 RPM
7 days' notice of changes	Before any model or price change, so labs and budgets can be adjusted in time
Use of Netra's name and logo	For the cohort name, class materials and certificates (ask 02)
A named Netra contact	One person for the pilot, the guest session and the reports

Support needs no separate channel: mentors use Netra's Discord for questions during the class.

09 · WHAT VETERANTECH COMMITS TO

What Netra gets back.

Product feedback from 8 live apps, public showcases and a path into client work. Behind those four commitments sits the work of running the class.

01 **Two usage and feedback reports** — WEEKS 4 AND 8

Who builds, where they got stuck, structured-output and tool-calling quality, and speed at peak load. Contents in section 10.

02 **Mentee apps anyone can try** — WEEK 8

Each team ships a live, open-source app built on Netra and credits it, like a hackathon gallery.

03 **A showcase that tags Netra** — AFTER DEMO DAY

Final projects featured on Veterantech's LinkedIn and social channels, with Netra tagged.

04 **Netra first for client AI work** — AFTER THE PILOT

When a Vexpro client project needs LLM inference, we propose Netra first. The client makes the final choice, so we promise no volume.

RUNNING THE CLASS

Cohort

Screening, team forming, attendance and the drop-out rule, with same-day key revocation.

Teaching

1–2 mentors per session, the curriculum, grading, and a deployment guide with extra consultation hours before week 7.

Starter kits

Backend with gateway client and streaming, golden-set runner, agent and MCP boilerplate, RAG ingest and eval pipeline.

Gateway

Keys, rate limits, revocation, request logs and database backups for the length of the pilot.

Key security

The Netra key lives only in the gateway's server environment, never in a repo or a mentee's hands.

Privacy

No names or student numbers in gateway data or reports. Reports show counts and patterns, not individuals.

10 · REPORTING AND TIMELINE

Two reports, one review, one demo day.

Netra already sees speed and errors on its own servers. The reports add what it cannot see from there: who builds, where new developers get stuck, and how the service performs from the user's side.

AREA	WHAT WE REPORT	SOURCE	INTERIM WEEK 4	FINAL WEEK 8
Adoption	Weekly active builders and tokens used	Gateway logs	●	●
	Apps live by demo day	Public repos	○	●
Feedback	Where new developers get stuck: docs, errors, SDK gaps, ease of integration	Mentor log	●	●
	Structured-output and function-calling quality	Labs and assignments	●	●
Speed	Time to first token and tokens/sec, p50 and p95	Gateway logs	●	●
	Error rate on Netra's side, by model and hour	Gateway logs	●	●
	Performance at peak load in live sessions	Gateway logs	●	●

Speed is measured at our gateway server, so it includes the network between us and Netra. Time to first token counts streamed calls only. Error rate leaves out the gateway's own rejections (422, 429, 401, 403).

TIMELINE

WEEK 0	WEEK 1	WEEK 2	WEEK 4	WEEK 8	NEXT
Starting pool opened. Netra key wired into the gateway and tested. Selection closes, keys issued.	Kickoff. Every mentee connected through the gateway.	Pool for weeks 3–8 sized together from real usage.	Interim report and review with Netra: adjust the pool, models or scope.	Demo day with a Netra judge. Final report.	Decide together on a second cohort or a hackathon.

Report format. A short written report with the numbers above, sent to the Netra contact after weeks 4 and 8 and walked through on a call.

Dates. Calendar dates for every step follow once both sides agree the start date. Weekly numbers stay available in the gateway logs on request.

11 · SCOPE AND OPEN POINTS

What this pilot covers, and what it does not.

A narrow scope keeps the pilot easy to judge at week 8. Anything outside it can be part of a next step.

IN SCOPE

- One 8-week cohort of the AI Engineer class
- Netra credits for class work only, via the gateway
- Two reports and the week-4 review
- Demo day, the public app gallery and the showcase
- A Netra guest session and a judge's seat at demo day
- The cohort run as "powered by Netra", if Netra approves

NOT IN THIS PILOT

- A recruitment or talent pipeline. No candidate lists or student data go to Netra.
- Course materials licensed for Netra to reuse
- A dedicated support line. Netra's Discord covers questions.
- A money value for the credits before prices are confirmed

TO CONFIRM BEFORE WEEK 1

ITEM	OWNER	STATUS
Start date and session schedule	Both	OPEN
Models with tool calling and embeddings, and their prices	Netra	OPEN
Starting pool, then the pool sized after week 2	Both	OPEN
Bursts up to 20 RPM; 7 days' notice of model or price changes	Netra	OPEN
Per-key budget and rate-limit levels for the class	Both	OPEN
Netra's name and logo on the cohort and certificates	Netra	OPEN
Do per-key budgets in Netra's dashboard hard-block?	Netra	QUESTION
Guest session date; demo day online or on site	Both	OPEN
Planned gateway controls (page 07)	Veterantech	PLANNED

Assumptions. Sessions run online on Zoom. Netra shares the setup details on page 09 before week 1. Changes to the pool, models or scope go through the week-4 review or a written note agreed by both sides.

12 · AGREEMENT

Three yeses start the first cohort.

Credits and models, your name on the cohort, and Netra on stage. Once those are agreed, the gateway switches from the mock model and week 1 can be scheduled.

01

Confirm the starting pool and models

Netra shares the class key, base URL, model IDs and prices.

02

Wire the gateway

We switch from mock to Netra, test on a small budget and issue mentee keys.

03

Kick off week 1

On a start date agreed by both sides, with the Netra contact named.

FOR KSM VETERANTECH

NAME [Full name]

TITLE [Title]

SIGNATURE

DATE [DD Month 2026]

FOR NETRA

NAME [Full name]

TITLE [Title]

SIGNATURE

DATE [DD Month 2026]

× Netra Runtime

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